Logical Foundations of Concurrent Computation

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A brief introduction to **session types** for concurrency, and to the logical foundations of message-passing computation.

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- In a nutshell, a **session** provides a structure for a series of related interactions between communicating programs.
- A disciplined usage of sessions is key to ensure program correctness.
  Not only: sessions have elegant **logical foundations**!
- **Linear Logic**: A resource-oriented view on propositions / session protocols.
  Linear resources can be used exactly once.

Concurrent Programs

Message-Passing Concurrent Programs

- Software components (**services**) distributed across networks

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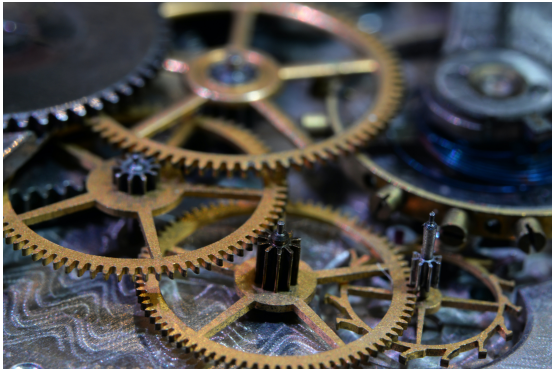
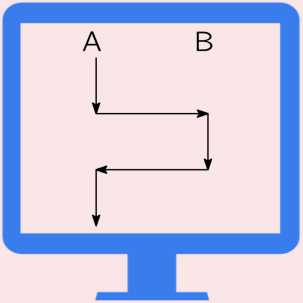
An (imperfect) analogy:



When is a Program Correct?

Sequential Programs



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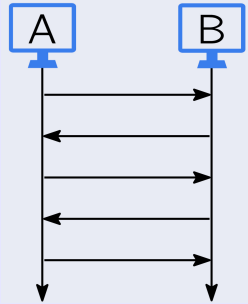

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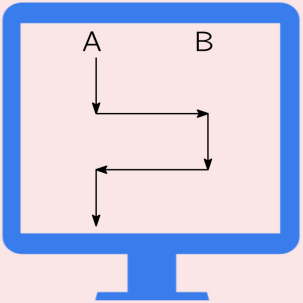

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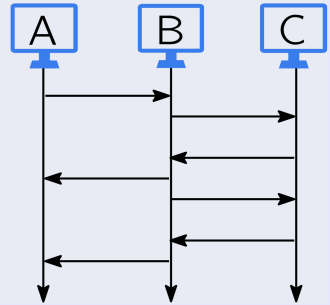

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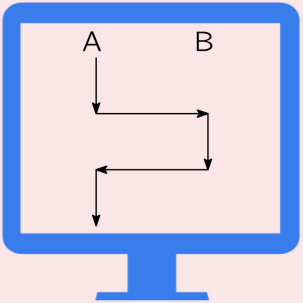

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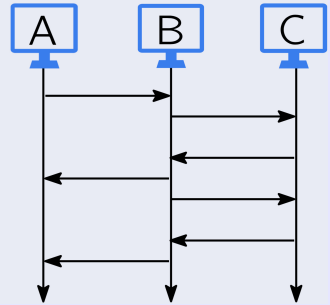

"Programs always respect their intended **protocols**"

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Note: The structure of messaging matters!

Example: A Two-Buyer Protocol

Desiderata for the implementations of Alice, Bob, and Seller:

- **Fidelity** – they **follow the intended protocol**.
  - Alice never ask Bob twice within the same conversation
  - Alice doesn't continue the transaction if Bob can't contribute
  - Alice chooses among the options provided by Seller

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- **Fidelity** – they **follow the intended protocol**.
- **Safety** – they don't feature **communication errors**.
  - Seller always returns an integer when Alice requests a quote

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- **Deadlock freedom** – they do not **"get stuck"** while running the protocol.
  - Alice eventually receives an answer from Bob on his share.

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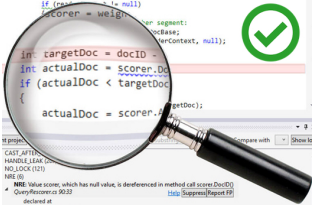
Correctness

- A **non-trivial notion**, which follows from the interplay of these properties.
- **Hard to enforce** due to actions being "scattered around" in programs.

→ A session specifies a protocol's structure, enabling program verification. It stipulates **what** and **when** should be exchanged (along a channel)

Type Systems

- Can detect bugs before programs are run
- Attached to many programming languages
- Implement a specific notion of correctness
  A program is either correct or incorrect



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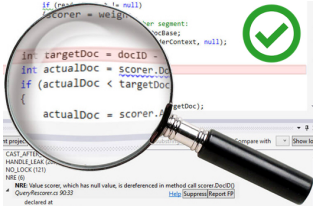

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- **Data** type systems classify values in a program
- Examples: Integers, strings, etc

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Concurrent Languages

- **Behavioral** type systems classify protocols in a program
- Example: “first send username, then receive true/false, finally close”
- A typical bug: sending messages in the wrong order

Protocols as Session Types

Session types specify protocols in terms of

- communication actions: send and receive
- choices: offers and selections
- recursion

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Session protocols are attached to **interaction devices**:

- communication channels in programs (think Go and Rust)
- TCP-IP sockets
- ...

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A formal syntax for protocols:

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Notice:

- Sequential communication patterns (no built-in concurrency)
- U stands for basic values (e.g. `int`) but also other sessions S

The Two-Buyer Protocol, Revisited

Recall the protocol between Alice, Bob, and Seller:

1. Alice sends a book title to Seller, who sends a quote back.
2. Alice checks whether Bob can contribute in buying the book.
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Session Types for The Two-Buyer Protocol

Two independent protocols, with Alice "leading" the interactions:

1. A session type for Seller (in its interaction with Alice):

$$S_{\mathsf{SA}} = ?\mathtt{book};\ !\mathtt{quote};\ \&\begin{cases}\text{buy}: & ?\mathtt{paym};\ ?\mathtt{address};\ !\mathtt{ok};\ \mathsf{end}\\ \text{cancel}: & ?\mathtt{thanks};\ !\mathtt{bye};\ \mathsf{end}\end{cases}$$

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Duality relates session types with opposite behaviors:

- The dual of sending is receiving (and vice versa)
- Branching is the dual of selection (and vice versa)

The dual of session type S is written $\overline{S}$.

Session Types for The Two-Buyer Protocol

Example:

- Recall that S_{AB} describes Bob's viewpoint in her interaction with Alice:

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- That is, Alice's implementation should conform to **both** $\overline{S_{\mathsf{SA}}}$ and $\overline{S_{\mathsf{AB}}}$.

Session Type Duality, Formally

Given a (finite) session type S, its dual type $\overline{S}$ is inductively defined as follows:

$$
\begin{aligned}
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\overline{\mathsf{end}} &= \mathsf{end}
\end{aligned}
$$

Notice:

- For recursive types, duality is defined **coinductively** (the dual of $\mu t.S$ is *not* $\mu t.\overline{S}$)

Propositions as Sessions

A Curry-Howard correspondence for concurrency:

Linear logic propositions	$\leftrightarrow$	types describing **behavior** (sessions)
Sequent calculus derivations	$\leftrightarrow$	**concurrent processes** (π-calculus)
Cut reductions	$\leftrightarrow$	communication between processes

Key Ideas

- Assume a language of **processes** (denoted $P, Q, \ldots$) that interact by synchronizing on **names** (denoted $x, y, z, \ldots$).

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- Processes can send/receive messages on names, using sequencing and parallel composition. We "extract" their syntax from proofs.

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- Interpret the logical sequent

$$A_1, \cdots, A_n \vdash C$$

as a **typing judgment**, under a suitable reading of propositions as sessions:

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... by relying on sessions $A_1, \ldots, A_n$ on channels $x_1, \ldots, x_n$

Linear Logic Propositions as Session Types

`end`	**terminate** the session	1
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- Logical foundations (cut elimination) endows types processes with key properties, including deadlock freedom.
- Open challenges:
  - Stronger forms of deadlock and lock freedom by typing
  - Handling non-linear (shared) resources beyond ! (bang) in Linear Logic
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  - ♡ **Coinductive session protocols and connections with proof assistants**: Our national consortium "Cyclic Structures in Programs and Proofs".